

Autonomous Hand Gesture Recognition Solution for Embedded Automotive Systems

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ABSTRACT

We investigated how to design a software pipeline to recognize hand gesture from images inside a vehicle cockpit. Two architectures are compared. Frequency descriptors derived from the Fourier transforms were solely used in image outline analysis.

Index Terms: Hand gesture recognition, Pose estimation, Convolutional Neural Networks (CNN), Image segmentation, Classification, Automotive.

1 INTRODUCTION

"Contactless" gesture recognition promise to make vehicles safer while enhancing user experience. Gesture recognition allows intuitive manipulation of car's functionality without button and tactile screen. High-end models of car manufacturer already implement "contactless" gesture control, in BMW 7 Series in 2015 for instance, to control certain vehicle functions. However, car's manufacturer constraints linked to effectiveness under variable conditions such as light changes, user flexibility, accuracy, low-latency, computer resources, electric consumption challenge the extension of such systems to a mainstream audience.

The main contributions of the paper are the following. We investigate the performance of a software pipeline independent of any third-party solutions like Mediapipe Hands [1] able to cope with these constraints. For this, we build our own database of hand posture images. This permits to train two comparative networks (U-Net [2] and YOLO [3, 4]) for image segmentation and silhouette outline. From the hand contour, we create a polygon, combining *Ramer-Douglas-Peucker* algorithm, and B-Spline interpolation. We compute the Fourier descriptors from the points of this polygon, which feed a classifier.

2 DESCRIPTION OF THE SYSTEM

2.1 Automated database acquisition and labeling

A Raspberry Pi 3B+-based prototype was developed to acquire new user gesture from the cockpit without need of an additional PC. Semi-automated labelling of captured images was performed with a Computer Vision Annotation Tool. We acquired a large database of 54305 images, under various weather (sunny, cloudy,

partly cloudy), times of the day (sunrise, morning, afternoon, sunset, night) and ceiling light (off, on) conditions. A unique camera point of view was optimised to capture the pinch gesture.

2.2 Image segmentation and silhouette outline

A mandatory step before computing any contour descriptor is hand segmentation. We predict that this segmentation step is more time-consuming than the computation of the descriptors. Hence we pay a particular attention to the computational load of this segmentation step. Two convolutional neural networks (CNN) will be compared in terms of accuracy and computational time either on CPU or GPU processors.

2.3 Descriptor construction and classification

Out of the segmented hand, the pixel coordinates of the contour are transformed into a polygon with Ramer-Douglas-Peucker algorithm and then re-sampled with B-Splines to standardize the number of polygon points.

This algorithm yields a list of ordered points, from which we compute Fourier descriptors: let N be the number of points on the hand contour, and (x_n, y_n) their coordinates. Each point is associated with a complex number $z_n = x_n + iy_n$ ($i = \sqrt{-1}$, and $n = 0, \dots, N-1$). The discrete Fourier transform applied to this sequence of points is computed as follows:

$$Z_k = \sum_{n=0}^{N-1} z_n \cdot e^{-\frac{i2\pi kn}{N}}, k = 0, \dots, N-1 \quad (1)$$

where k is the frequency index of any coefficient (that we also call 'descriptor'), and Z_k the corresponding descriptor. The coefficients associated to the low frequencies capture the general structure and aspect of the hand, while those associated to the high frequencies reveal finer details. A good compromise between computational load and detail preservation led us to retain 18 coefficients. Removing the phase ensures the invariance to rotation; removing the zero-valued frequency, ensures the invariance to translation, and scaling the magnitudes ensures the scale invariance. The zero-frequency component may be used to locate the center of mass of the hand in the image. As for the classification method, we selected support vector machine (SVM) with a radial basis function (RBF) kernel, preceded by dimension reduction with principal component analysis (PCA), which preserves as much variance as possible in the reduced data.

3 RESULTS

We showed that despite U-Net accurately predicts the shape of segmented objects but also classifies efficiently pixels regarding of present/absence of object, YOLO outperformed U-Net. However, the Dice coeff is closer to 1 with YOLO than U-Net (0.943 vs. 0.931), suggesting a slightly better segmentation performance.

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The Dice loss is lower with YOLO than U-Net (0.056 vs 0.068), suggesting a better prediction of shape. Inference time is mainly degraded by using CPU instead of GPU with U-Net (12.265 vs. 123.305 ms) while it is quite stable with YOLO-v8 nano (81.271 vs. 63.561 ms).

Fig. 1 illustrates the ability of the proposed method to differentiate between two close postures of 'pinch'. The polygon points obtained when the fingers touch each other are completely different from those obtained when the fingers are only close to each other, but not touching. On the contrary, the landmarks provided by the mediapipe AI toolbox [1] are very similar from one case to another.

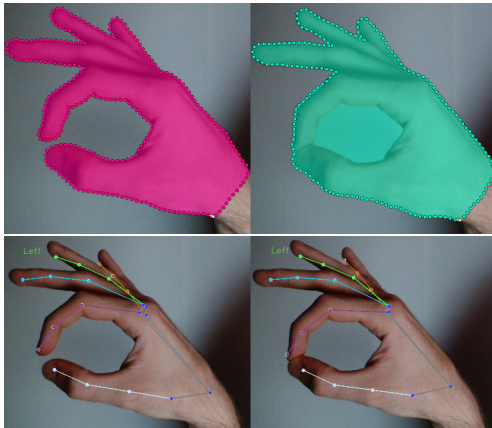


Figure 1: Comparative exemplification of landmark extraction: proposed method (up), and comparative mediapipe method (down). The differentiated annotation of two close postures of pinch permits the class separation with a fast SVM classifier.

Fig. 2 represents 7 classes in the 3D space, through the 3 main PCA components. We showed that an optimal dimensionality in vectorial subspace of PCA used to classify descriptors with SVM by RBF kernels was obtained with PCA (4), that displayed a maximal accuracy (100%) and the second shortest inference time (12,452 μ s).

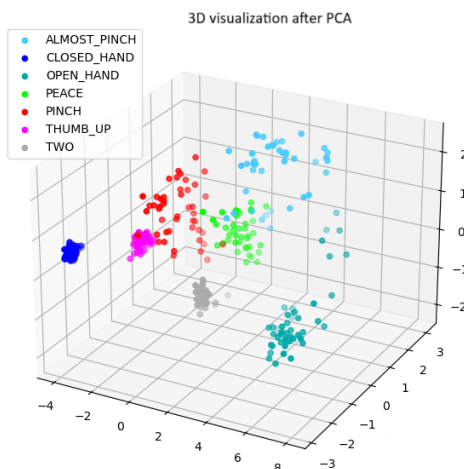


Figure 2: 3D scatter of the three main PCA components of the descriptors: Notice that 'almost pinch' and 'pinch' classes are well separated.

Fig. 3 is extracted from this video and displays the original RGB

image, the segmented hand (with U-Net), and its use in an interface implemented with Unity.



Figure 3: Demo: hand segmentation and use in interface.

4 CONCLUSION

We propose to adapt and combine adequately a CNN for hand segmentation, the Ramer-Douglas-Peucker algorithm for contour polygonisation, and B-Splines for resampling and standardization of contour landmarks. From these landmarks, we compute descriptors which are invariant to translation, rotation, and scaling, based on complex fourier coefficients. This permits an accurate distinction between two postures which are 'almost pinch' and 'pinch', compared to the gold standard for hand segmentation, namely 'mediapipe' toolbox.

For this, the required segmentation performed by the CNN has been carefully optimized in terms of software and hardware: we selected the best CNN model (comparing U-Net and YOLO) and the best hardware architecture (between CPU and GPU) in terms of segmentation accuracy and computational time of the inference phase. The lower dependence on GPU acceleration and relatively stable temporal performance on CPUs of YOLO-v8 nano model stands out as the optimal trade-off between efficiency and performance for systems with limited hardware resources. One of the main objectives of these descriptors is clear separability of gesture classes related to specific cognitive actions and the perception of touch by the user. Indeed, the joining or disjoining of shapes described by the frequency-based descriptors is closely correlated with nervous perception, particularly in the context of tactile or gestural interaction. This plays a key role in posture identification: the recognition of gestures such as pinching, pressing, or finger bending is directly influenced by how frequencies are distributed along the hand contour. In the future, complementary shape criteria (e.g., geometric moments or descriptors based on elongation or compactness) could complete the analysis to capture more complex geometric aspects of objects and enhance the accuracy of classification models.

SUPPLEMENTAL MATERIALS

A video is available at : <https://amubox.univ-amu.fr/s/3P2xtD8Q6WeRATE>.

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